

Capacitor Charging and the Time Constant

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September 4, 2026

Abstract

This document introduces the charging behaviour of a resistor-capacitor (RC) network and derives the capacitor voltage at one time constant.

1 Introduction

When a capacitor C is charged through a series resistor R from a constant supply voltage V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right).$$

The product RC is called the time constant,

$$\tau = RC.$$

2 Voltage at the Time Constant

Substituting $t = \tau$ into the charging equation,

$$V(\tau) = V_0 \left(1 - e^{-\tau/RC}\right)$$

Since $\tau = RC$,

$$V(\tau) = V_0 (1 - e^{-1})$$

$$V(\tau) \approx V_0(1 - 0.368)$$

$$V(\tau) \approx 0.632V_0$$

For $V_0 = 5.0$ V,

$$V(\tau) \approx 0.632 \times 5.0 \approx 3.16 \text{ V}$$

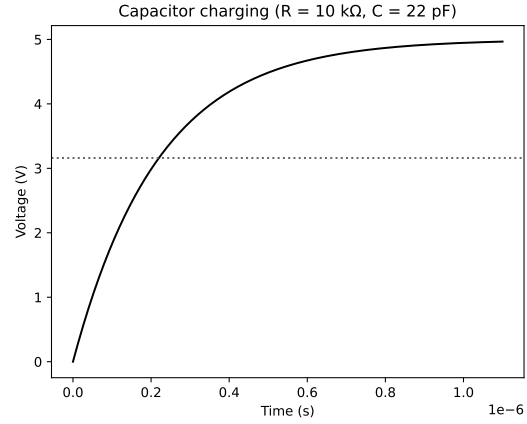


Figure 1: Charging curve for $R = 10 \text{ k}\Omega$, $C = 22 \text{ pF}$.